

Oleksii Odarchuk

Backend Developer (Go) · Creator of rombik and the Piton language



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Performance-obsessed engineer with a product mindset. 1st place at BEST Hackathon 2026, 3rd at Mate Hackathon, invited to DOU Day 2026 as a Student with High Potential. Builds real products fast — from hackathon MVPs to production Go services.

WINS & HACKATHONS

1st Place — Lead Backend & Sensor Fusion April 18–19, 2026

BEST Hackathon 2026 — GPS-less Drone Trajectory

- Algorithm to **reconstruct a UAV's trajectory without GPS** — from IMU, barometer and other inertial-sensor logs. Sensor fusion in **Go + C**: Go for the pipeline & integration, C inline for the hot numerical kernels.
- 1st place among top teams** — judges singled out the architectural resilience: modular split telemetry / integration / simulation let us swap datasets without rewriting.

3rd Place — Backend Engineer March 14, 2026

Mate Hackathon (50 participants, 10 teams)

- Designed and built a Go backend from scratch in **7 hours** for an AI-powered MarTech tool that automates competitor-ad analysis.
- Dual AI pipeline: GPT-4o for marketing trigger analysis + Nano Banano 2 (fal.ai) for creative generation; backend routed all AI calls to cloud APIs, sustaining throughput under hackathon load.
- Beat teams with 5+ years of experience by focusing on a stable MVP — **3rd of 10 teams**.

Security Researcher → Job Offer 2026

College Blockchain System "Student Hryvnia"

- Discovered and exploited a critical vulnerability: frontend capped transaction amounts, **backend had no such check** — enabling arbitrary negative-value transfers. Reported responsibly; received a job offer from administration.
- Reverse-engineered the official JS miner through Webpack bundles — then built a faster Go replacement with a goroutine worker pool (SHMiner — p. 2).

DOU Day 2026 — Student with High Potential

Personally invited by the organisers with a free ticket — a dedicated track for promising student developers.

TECH STACK

Languages: **Go** **Python** **C**

TypeScript

Backend: **Gin** **Fiber** **REST**

WebSockets **Goroutines** **JWT**

OAuth 2.0

Databases: **PostgreSQL** **Redis**

ClickHouse

Frontend: **Svelte** **Wails**

Tailwind

AI / APIs: **GPT-4o** **Fal.AI**

DevOps: **Docker** **Linux (Arch)**

Git **TinyGo** **Bruno** **Typst**

EDUCATION & LANGUAGES

Network Technologies — IT

College NULP, 2nd year
2024 — present

IT Class — Science Lyceum, Rivne
2022 — 2024

Ukrainian — native · **English** — B1

PROJECTS

◆ **rombik — Flowchart Generator (SaaS)**

[website](#) → [API](#) →

My first full-fledged commercial SaaS · code in 6 languages → a DSTU 19.701-90 flowchart

Designed and run entirely by me. A **Go engine** — a compiler pipeline (parser → IR → layout → render) on **tree-sitter** for **Python, C, C++, C#, Java, Pascal**; pixel-for-pixel to DSTU, byte-exact golden tests. Web editor (pan/zoom, lasso-splitting), 6 formats (SVG/PNG/PDF/DOCX/Typst/Excalidraw). **Billing runs on my own go-monobank-sdk** (HMAC webhooks, grants, receipts). Compiler craft carried over from my language **Piton**. Public **API, CLI, MCP server**, an AI skill and describe→flowchart via AI. Free preview, export needs an account.

Go · Gin

tree-sitter

API · CLI · MCP

go-monobank-sdk

PostgreSQL

DSTU

OPEN SOURCE SDK

go-monobank-sdk — [github](#) → **Go SDK for every monobank API**

Full coverage of **personal**, **provider** (with monoKEП), **business** for legal entities, **acquiring** (subscriptions, monopay button, split-receivers, T2P) and **Installment Purchases**. ECDSA webhook verification with auto-rotating keys, drop-in `http.Handler`, typed money/currency/MCC, HMAC-SHA256 for installment callbacks, a fake monobank server for tests, and a separate OpenTelemetry sub-module.

Go ECDSA

HMAC-SHA256

OpenTelemetry iter.Seq2

SDK CONSUMER

monokasa — Telegram [github](#) → **Ticket-Selling Bot**

Sells tickets to a single show through a monobank jar and issues PDFs with QR codes scanned at the venue entrance. Live consumer of my own **go-monobank-sdk**: webhook auto-registers at startup, reservations get a 15-minute hold, and show-time reminders fire at-most-once even across restarts.

Go SQLite telebot

PDF + QR

go-monobank-sdk

1000× FASTER

SHMiner — Desktop [github](#) → **Crypto Miner**

Cross-platform desktop miner built with **Wails (Go + Svelte/TypeScript)**. Rewrote SHA-256 hashing from JS to Go with a goroutine worker pool — achieved **1000× faster hashrate** than the official client on identical hardware. AES-encrypted wallet storage, real-time WebSocket dashboard, multi-wallet, zen mode.

Go Svelte TypeScript

Wails WebSockets

Goroutines

150+ USERS

AbitAssistant Bot [github](#) →

Telegram bot helping Ukrainian students track university admission rankings. Reverse-engineered closed third-party APIs to fetch live competition data. **150+ active users in first year**, marketing campaign planned for 2026.

Python Aiogram

PostgreSQL BeautifulSoup

Pandas Docker

LinkShortener Bot [github](#) →

Telegram URL shortener with QR generation and geo analytics. PostgreSQL for storage, Redis for redirect caching, ClickHouse for real-time click analytics (country, city, platform). Docker, GeolP2.

Go PostgreSQL Redis

ClickHouse Docker

Telegram API

MY OWN LANGUAGE

Piton — My own [github](#) → [docs](#) → **programming language**

Ukrainian transliterated language + interpreter, from scratch in **Go without yacc/bison**: recursive descent parser, tree-walking evaluator, Turing-complete. Flowcharts via a built-in **D2**-based engine. Interpreter compiled to **WebAssembly via TinyGo** (220 KB).

Go TinyGo WebAssembly

D2 Nix